

Shearwaters: 10% of the world population?

*Testing BirdLife Malta's figure for the Yelkouan shearwater
against regulator and published estimates*

THE CLAIM UNDER REVIEW

“Malta’s population is estimated to be around 1,600 – 1,800 pairs, constituting approximately 10% of the global population.”

BirdLife Malta, LIFE Arcipelagu Garnija project page (undated; read 5 October 2026).
On the Yelkouan shearwater (Garnija), classed as Vulnerable by the IUCN.

VERDICT: LARGELY SUPPORTED

An important share; “about 10%” rounds up a central 6–8%

Version 1.0 · 5 October 2026

Status: no right of reply needed

Prepared from public sources and published literature.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

On its page for the LIFE Arċipelagu Garnija project, BirdLife Malta writes that the Yelkouan shearwater (*Garnija*) “**Malta’s population is estimated to be around 1,600 – 1,800 pairs, constituting approximately 10% of the global population.**” We compared both numbers with the regulator’s estimate and the published global estimates.

- 1** **Malta’s count is plausible but older than the regulator’s.** The Environment and Resources Authority’s Species Action Plan 2022–2030 gives 1,795–2,635 breeding pairs, overlapping and above BirdLife Malta’s 1,600–1,800. The BirdLife page is undated; its project ran 2016–2020.
- 2** **The global population is poorly known.** A peer-reviewed review (Bourgeois & Vidal 2008, *Oryx*) put it at 11,355–54,524 pairs, adding that most censuses are probably overestimates. The figure most often quoted today is 21,000–36,000 pairs (CIESM guide).
- 3** **“About 10%” is at the top of the range.** With BirdLife Malta’s own count and 21,000–36,000 global pairs, Malta’s share is 4.4–8.6% (central 6%). With the regulator’s count it is 5.0–12.5% (central 7.8%). The regulator also says “around 10%”.
- 4** **Verdict: largely supported (moderate confidence).** Malta holds an internationally important share of a vulnerable seabird, which is the point of the sentence. The precise share cannot be pinned down, and the central estimates are closer to 6–8% than 10%.

THIS CLAIM



1,600–1,800

Breeding pairs in Malta, BirdLife Malta (undated)

1,795–2,635

Breeding pairs in Malta, ERA Species Action Plan 2022–2030

4–13%

Malta’s share with either count and 21,000–36,000 global pairs

6–8%

Central estimates of the share (claim: about 10%)

WHAT WOULD MOVE THE VERDICT UP

A current, sourced global estimate under about 20,000 pairs, or a Maltese count near the top of the regulator’s range, either of which would put Malta at about 10%.

WHAT WOULD MOVE IT DOWN

A current global estimate well above 36,000 pairs, or a new Maltese census below 1,600 pairs; then “about 10%” would overstate Malta’s share.

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5 Testing the claim

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7 Limitations

1

The claim and what we could verify

The sentence is on BirdLife Malta’s page for LIFE Arcipelagu Garnija (LIFE14 NAT/MT/000991, 2016–2020) [1], read on 5 October 2026; the page carries no date. The same text appears on the LIFE PanPuffinus project’s species page. The European Commission’s LIFE project page [2] describes the project’s aims (to increase breeding pairs by about 10% and reproductive output by 25%) but gives no population total.

Source	What it says	Our access	Status
BirdLife Malta, LIFE Arcipelagu Garnija page [1]	“Malta’s population is estimated to be around 1,600 – 1,800 pairs, constituting approximately 10% of the global population.”	Read, 5 Oct 2026.	The claim
ERA, Species Action Plan 2022–2030 page [3]	“around 10% of the global breeding population ... an estimated population of 1,795–2,635 breeding pairs”.	Internet Archive copy, 8 Aug 2025 (live site 403).	Regulator
Bourgeois & Vidal (2008), <i>Oryx</i> [4]	Global population 11,355–54,524 pairs; probably overestimated.	Full text read (free to read).	Peer-reviewed
CIESM seabird guide [5]	Global 21,000–36,000 pairs; 46,000–92,000 individuals.	Read; no source given ♦.	Context

2

Method

Question. Is Malta’s population about 1,600–1,800 pairs, and is that about 10% of the world’s?

Evidence. We searched the peer-reviewed literature (OpenAlex, Crossref) and official sources for Maltese and global estimates, recorded them in *data/cc-106/population_estimates.csv*, and computed Malta’s share for each combination (*tools/cc-106-report/calc.py*): the lowest Maltese estimate over the highest global one, the reverse, and the ratio of midpoints. The IUCN Red List and BirdLife International data pages refused automated access, and we could not query the EU Birds Directive (Article 12) reports directly, so the current official global estimate is known to us only through the CIESM guide.

Grades. Bourgeois & Vidal (2008): C (review). ERA plan: C (official). CIESM guide: C, second-hand. **Verdicts** follow the five-point scale in Appendix A.

3

What the evidence shows

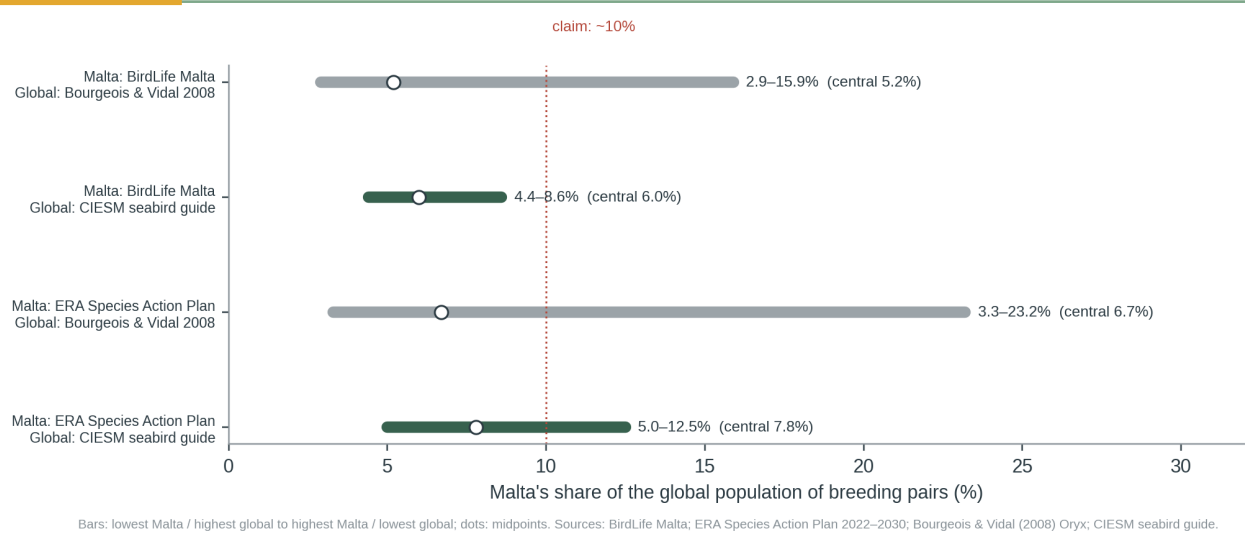


Figure 1. Malta’s share of the world’s Yelkouan shearwater breeding pairs under each pair of estimates. Green: with the 21,000–36,000 global estimate.

With the counts BirdLife Malta states, Malta’s share cannot exceed 8.6% unless the global population is below about 21,000 pairs. With the regulator’s newer count, 10% falls inside the range. The 2008 review’s range is so wide that it supports anything from 3% to 23%. A study of the largest colony, Tavolara in Sardinia, estimates it holds more than half of the world population (Pezzo et al. 2021) [6].

4

Where the evidence points different ways

Is “approximately 10%” fair?		AT THE TOP OF THE RANGE
FOR The regulator uses the same figure with a higher Maltese count, and 10% lies inside the range for that count (5–12.5%). The global total is uncertain enough that 10% cannot be ruled out.	AGAINST The page’s own count, with the most-quoted global estimate, gives 4.4–8.6%. Central estimates are 6–8%. No source we found gives a calculation behind “10%”.	
WHY THEY DIFFER, AND WHAT IT MEANS HERE The share is of the right order and Malta’s population is internationally important; “about 10%” rounds up. A safer wording would be “roughly 5–10%”.		

5 Testing the claim

Sub-claim	Said by	What the evidence shows	Rating
A. Malta's population is around 1,600–1,800 pairs	BirdLife Malta [1]	Overlaps the regulator's 1,795–2,635 pairs (2022–2030 plan); the page is undated and its count is the lower.	PLAUSIBLE, DATED
B. ...approximately 10% of the global population	BirdLife Malta [1]	4.4–8.6% with the page's own count; 5.0–12.5% with the regulator's; central 6–8%. Global total poorly known.	UPPER END OF RANGE

6 Verdict and requests for evidence

VERDICT

LARGELY SUPPORTED

Malta holds an important share; “about 10%” rounds up a share closer to 6–8%. Confidence: moderate.

Why. Both numbers are of the right order and consistent with official and scientific sources, but the share is at the upper end of what the estimates allow and the global total is poorly known. Confidence is moderate because we could not read the current IUCN and Article 12 figures directly. **What this verdict does not say.** It does not question the species' conservation need or BirdLife Malta's conservation work; it tests a population share.

Evidence we are asking for

- ☐ From BirdLife Malta: the year and source of the 1,600–1,800 pairs, and the global figure behind “10%”.
- ☐ From ERA: the method behind 1,795–2,635 pairs and Malta's latest Article 12 report figures.

RIGHT OF REPLY

Not needed for this verdict (maintainer rule of 5 October 2026).

7 Limitations

- The IUCN Red List and BirdLife International datazone refused automated access; the 21,000–36,000 figure is from the CIESM guide, which gives no source (♦).
- Seabird colonies in cliffs and caves are hard to census; all estimates are ranges with wide uncertainty.
- The ERA plan page was read from an Internet Archive copy; the plan document itself was not read.

References

- [1] BirdLife Malta (undated). LIFE Arcipelagu Garnija. Read 5 Oct 2026. <https://birdlifemalta.org/arcipelagugarnija/>
- [2] European Commission, CINEA. LIFE14 NAT/MT/000991 LIFE Arcipelagu Garnija, project page. <https://webgate.ec.europa.eu/life/publicWebsite/project/LIFE14-NAT-MT-000991/life-arcipelagu-garnija---securing-the-maltese-islands-for-the-yelkouan-shearwater-puffinus-yelkouan>
- [3] Environment and Resources Authority. Malta Yelkouan Shearwater Species Action Plan 2022–2030 (page). Internet Archive copy of 8 Aug 2025. <https://era.org.mt/malta-yelkouan-shearwater-species-ap-2022-2030/>
- [4] Bourgeois K. & Vidal E. (2008). The endemic Mediterranean yelkouan shearwater *Puffinus yelkouan*: distribution, threats and a plea for more data. *Oryx* 42(2):187–194. doi:10.1017/S0030605308006467. <https://doi.org/10.1017/S0030605308006467>
- [5] CIESM. Mediterranean seabird guide: Yelkouan shearwater. ♦ <https://ciesm.org/seabird-guide/yelkouan-shearwater/>

- [6] Pezzo F. et al. (2021). Productivity changes in the Mediterranean Sea drive foraging movements of yelkouan shearwater *Puffinus yelkouan*. *Marine Ecology* 42:e12668. doi:10.1111/maec.12668. (Abstract read.) <https://doi.org/10.1111/maec.12668>
- [7] MiŻien. Estimates and calculation: data/cc-106/; tools/cc-106-report/calc.py.

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	5 Oct 2026	First issue. Verdict Largely supported (moderate confidence); no right of reply needed.